MIL-T-83351A(USAF)
7 June 1973
SUPERSEDING
MIL-T-83351 (USAF)
12 January 1972

MILITARY SPECIFICATION

TRUCK, LIFT, AERIAL STORES MHU-124/E; 3,000 POUND CAPACITY, ELECTRIC MOTOR DRIVEN

1. SCOPE

1.1 This specification covers an electric motor driven, 3,000-pound capacity, self-propelled lift truck, designated MHU-124/E.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

Military

MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Requirements for

STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-162	Material Handling Equipment, Preparation for Shipment, Storage, Cyclic Maintenance, Outline Testing and Process
MIL-STD-209	Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-1273	Vehicles, Wheeled, Definition of Terms Related to

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:50Z
Check the source to verify that this is the current version before use.

DRAWING

Standard Manufacturing Company

5P1J Aerial Stores Lift Truck MHU-124/E

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 General requirements. The requirements of Drawing 5P1J (see 6.2) apply as requirements of this specification with the exceptions and additions specified herein.

3.3 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.5 Electromagnetic interference. The truck shall meet the requirements of MIL-STD-461, class C equipment.

3.6 Workmanship. The truck, including all parts and accessories, shall be constructed and finished in a workmanlike manner. Particular attention shall be given to the following:

a. Freedom from blemishes, defects, burrs, and sharp edges

b. Accuracy of dimensions, radii of fillets, and marking of parts and assemblies

c. Thoroughness of finishing, welding, painting, machining, and soldering

d. Alignment of parts and class fit of assembly, screws, and bolts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:50Z
Check the source to verify that this is the current version before use.

3.6.1 Cleaning. The truck shall be thoroughly cleaned of loose, spattered, or excess slag, metal chips, rust, oil, grease, and other foreign matter prior to subsequent production line operations and after assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection and testing requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the truck shall be classified as follows:

a. Preproduction testing . . . . . See 4.4

b. Quality conformance tests . . . . . See 4.5.

4.3 Test conditions

4.3.1 Standard test conditions. Unless otherwise specified herein, all tests shall be conducted under normal ambient room temperature and atmospheric conditions.

4.3.2 Test load. A dead-weight test load of 3,000 pounds shall be used to simulate the capacity and configuration of loads that will be handled during operational use. The test load shall be a cylindrical tube with an approximate length and diameter of 140 and 18 inches, respectively.

4.3.3 Data. During the testing specified herein, data shall be recorded at intervals to insure compliance of the truck with the performance requirements specified. Recorded data shall include the date, time, and ambient temperature at which the test is conducted as well as the data needed to accurately calculate the performance rates, efficiencies, et cetera, to be determined by the specified tests.

4.3.4 Failure during test. Characteristics exhibited by the truck during any test that would cause the truck to fail any requirement specified herein while in service shall constitute failure of the unit to pass the test and shall be the basis for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:50Z
Check the source to verify that this is the current version before use.

4.4 Preproduction testing (see 6.2)

4.4.1 Test sample. One truck shall be subjected to the preproduction tests specified 4.4.3.

4.4.2 Test report. Upon completion of the preproduction tests, a test report shall be prepared in accordance with MIL-STD-831 and three copies of the report furnished to the procuring activity.

4.4.2.1 Reliability and maintainability information. The following information shall be included as an appendix to the test report:

a. All failures, servicing, adjustments, maintenance, and irregular functioning shall be identified by accumulated operating time, cycles, miles, or position in the test procedure, as appropriate. Test conditions at the time of the events identified shall be recorded.

b. A summary of the engineering analysis and of any tests conducted to determine assignable cause for any failure or irregular functioning.

c. A summary of the engineering analysis leading to any corrections made to design, construction, quality control, or other procedures, or leading to any corrections to be made or proposed to be made to production items. The summary shall also include an analysis of the predicted effectiveness of such corrections.

d. Clock time and man-hours required for each maintenance and servicing action taken during the tests. A brief description of the experience and qualifications of the personnel taking such actions.

e. Test activity or contractor comments on item features or requirements that, if modified, should improve the item.

f. Test activity or contractor comments on use or maintenance conditions to be avoided or cultivated to increase the reliability or useful life of the item.

g. Any of the above information that is already included in the test report body need not be repeated in the information required by this paragraph, but clear reference to the location of the data shall be included.

4.4.3 Preproduction tests. The preproduction tests shall consist of all the tests described under 4.6, conducted in the same order.

4.5 Quality conformance tests. Each truck shall be subjected to the following tests as described under 4.6:

a. Examination of product . . . . . See 4.6.1

b. Operational tests . . . . . See 4.6.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:50Z
Check the source to verify that this is the current version before use.

4.6 Test methods

4.6.1 Examination of product. The truck shall be examined to determine conformance to the requirements of this specification with respect to dimensions, quality of workmanship, materials, design and finishes, markings, and visible defects, and the functional operation of all components, assemblies, accessories, and controls. Particular attention shall be given to the lift mechanism and its power supply system, the routing and protection of component parts, hydrostatic transmission, the positioning adjustments, and the overall structural stability of the truck.

4.6.2 Operational tests. The truck shall be operated under the following conditions to insure that the various components and capabilities comply with the requirements specified herein. During the operational tests, all components shall be operated according to the instructions provided with the truck.

4.6.2.1 Functional test. The truck shall be functionally tested by placing the rated load into position on the lift table when the lift mechanism is in the retracted position. The load shall be mounted on the table with particular attention given to the compatibility of the lift table mountings. The truck shall be operated through the entire range of lift and lowering. All table controls shall be operated and tested for proper operation. The truck shall be driven at speeds varying from minimum to maximum, fore and aft, for a total of 1 mile on a hard level surface. During this time, the control pedal, steering, and brakes shall be tested for proper operation. Any operational irregularity, instability, deformation, or failure shall be cause for rejection of the truck.

4.6.2.2 Counterbalance valves. The truck shall be tested to insure that each counterbalance (safety) valve is set to hold 4,500 pounds on the lift table per each lift cylinder. The right lift cylinder shall be hydraulically disconnected so that only the left cylinder is supporting the lift system. The table shall be 30 ±2 inches from the ground (unloaded) and all table motions centered. A load of 4,500 ±100 pounds shall be placed on the centerline of the table within ±2 inches. The counterbalance valve shall hold the load for a period of 2 minutes. During this 2-minute period, there shall be no visual settling of the load. The lift arm control valve shall be held in the open-down position during this period. The test shall be repeated with the right cylinder connected and the left cylinder disconnected. The frame or front wheels of the unit may be blocked to remove load from the tires, if desired, to limit deflection. The rear of the unit may be restrained.

4.6.2.3 Static proof test. The truck lift mechanism shall be subjected to a static proof test with a 300 percent proof load of rated capacity (3,000 pounds). The lift table shall be approximately 48 inches above ground level

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:50Z
Check the source to verify that this is the current version before use.

with the table head in maximum longitudinal (FWD) and lateral position. A load of 9,000 pounds shall be placed on the centerline of the table. The truck shall hold the load for at least 3 minutes duration. The rear of the unit may be restrained. Evidence of yield in any of the components which could in any manner prevent the truck from meeting the operational requirements during service life shall be cause for rejection.

4.6.3 Binding. The truck shall be tested for binding in the lift mechanism. Under capacity load, the truck shall be positioned on an uneven surface approximately 3° from the horizontal, in at least four different directions to test that binding will not occur in the operation of the lift mechanism. Evidence of binding shall be cause for rejection of the truck.

4.6.4 Cycle tests. The truck shall be subjected to a cycle test which consists of raising and lowering a capacity load through the full lift range of the lifting mechanism for 250 cycles. Lift table adjustments shall be actuated, one adjustment to each five lifting cycles. Also, each five lifting cycles shall be alternately tested in the high speed range and the low speed range for the system. During the cycle tests, the truck shall be inspected for leakage. Fluid lines, fittings, pumps, and other components shall show no leakage. However, if such equipment can be tightened to stop leakage that may have occurred during a cycle test, such adjustments will be acceptable. At the end of this test, evidence of yield in any components that could in any manner prevent the truck from meeting the operational requirements during service life shall be cause for rejection.

4.6.5 Settling rate. After the cycle tests in 4.6.4, the settling rate of the lift mechanism shall be measured. The lift mechanism shall be raised to its maximum position and with rated load on the table, the truck shall be allowed to stand for 1 hour. A measurement of more than 0.500 inch per hour shall be cause for rejection of the truck.

4.6.6 Manual operation. The truck shall be tested for manual operation. With power off, the truck with rated load shall be manually raised and lowered three times. During these cycles, each lift table adjustment shall be tested to insure the back-up capability of the truck.

4.6.7 Mobility. While supporting the 3,000-pound capacity load and operating on its own power, the truck shall be tested for type I, class 2, group C mobility in accordance with MIL-M-8090, with the following exceptions:

4.6.7.1 Steering. The mechanical linkage with power assist steering mechanism of the rear wheels shall be tested to insure positive control of the truck at all speeds. Compliance with the turning radius of 14.0 ±0.10 feet from turn center to outside corner of unit (35° rear wheel cramp angle) shall be in accordance with MIL-STD-1273.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:50Z
Check the source to verify that this is the current version before use.

4.6.7.2 Control pedal. The truck shall be driven at speeds varying from minimum to maximum fore and aft, for a speed control pedal test of 1 hour duration. The test shall insure that the control pedal is properly positioned for easy access and movement. During this time, the truck shall stop 20 times and the operator's foot removed from the control pedal to verify that the control pedal cannot be inadvertently engaged without deliberate movement by the operator. Indication of truck motion without actual movement of the control pedal by the operator shall be reason that the truck has failed this test.

4.6.7.3 Service brakes. The service brakes shall operate smoothly and apply uniform braking action during the functional tests (4.6.2.1). The performance of the service brake capability shall be tested for holding the loaded truck on a 10-percent incline. The rear wheels shall not roll.

4.6.5.4 Parking brakes. The parking brakes shall operate satisfactorily during the functional tests (4.6.2.1). The performance of the parking brake capability shall be tested for holding a loaded (3,000 pounds) truck on a 10-percent incline. The rear wheels shall not roll.

4.6.7.5 Towing. The unloaded truck shall have the hydrostatic drive disengaged, and towed by another vehicle at a speed of 2 mph over a paved level surface for a distance of 1 mile to determine compliance with the towing provisions. The truck shall not be towed over the obstructions of the type I test course.

4.6.8 Environmental tests. The loaded truck shall be subjected to the following tests conducted in accordance with the specified procedures of MIL-STD-810:

4.6.8.1 High temperature. The truck shall be placed in a hot chamber, the temperature raised to 120°F and maintained for 6 hours. The chamber temperature shall be raised to 154°F within a period of 1 hour and the unit subjected to a soak period for at least 4 hours. The chamber temperature shall be adjusted to 104°F and maintained until temperature stabilization of the truck is reached. At 104°F, the truck shall be subjected to the cycle tests (4.6.4) for 30 minutes. The lift mechanism shall be manually cycled three times during this test to insure its operation. After 30 minutes, with rated load at maximum raised position, the hydraulic system shall be checked for internal leakage. The settling rate of the lift mechanism shall not exceed 0.500 inch per hour.

4.6.8.2 Low temperature. The truck shall be subjected to low temperature storage and operation in accordance with method 502, procedure I. The truck shall be placed in a cold chamber, the temperature lowered to -30°F and maintained for 3 days. The chamber temperature shall be lowered to -50°F and the

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:50Z
Check the source to verify that this is the current version before use.

unit subjected to a soak period of at least 1 day. The chamber temperature shall be adjusted to +20°F and maintained until temperature stabilization of the truck is reached. At +20°F, the truck shall be subjected to the cycle tests (4.6.4) for 30 minutes. The lift mechanism shall be manually cycled three times during this test to insure its operation. After 30 minutes, with rated load at maximum raised position, the hydraulic system shall be checked for internal leakage. The settling rate of the lift mechanism shall not exceed 0.500 inch per hour.

4.6.9 Power train drive. The truck shall be subjected to a power train drive test. The test shall be of 20 hours duration, 10 hours loaded with 3,000-pound simulated missile and 10 hours unloaded. The test shall be conducted by driving a vee pattern where the truck backs down one leg of the vee, turns and drives forward up the other leg. Failure to conform to specified requirements shall be cause for rejection of the truck.

4.6.10 Electromagnetic interference. The truck shall be subjected to radiated interference in accordance with MIL-STD-462, method RE02 to show compliance with the requirements of MIL-STD-461. Failure to meet these tests shall be cause for rejection of the truck.

4.6.11 Lifting and tiedown rings. The lifting and tiedown rings and associated structure shall be subjected to a proof load test of 300 percent of the gross weight of the truck for compliance with MIL-STD-209.

4.6.12 Reliability demonstration and test. Satisfactory completion of all tests specified herein shall be a demonstration of compliance with the quantitative reliability requirements.

4.6.13 Maintainability evaluation. The truck shall be inspected and evaluated from the standpoint of ease of maintenance, servicing, and operation. Particular attention shall be given maintenance with a minimum number of common tools, servicing, and operation by personnel wearing heavy gloves, and provisions made to prevent accumulation of dirt, water, etc, that may hinder servicing and operation.

4.6.13.1 Preventive maintenance, servicing, and unit replacement. The frequency of servicing required for the various components shall be determined by study of existing supplier's recommendations and by servicing required during the preproduction tests. Interferences or obstructions to servicing shall be noted. Any tendency of the truck to pick-up foreign material which interferes with servicing and operation shall be noted. Removal and installation of major components and assemblies to demonstrate ease of replacement shall be accomplished. Particular attention shall be given to time required, difficulties encountered, and the tools and skills required in conducting maintenance, servicing, and replacement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:50Z
Check the source to verify that this is the current version before use.

4.7 Inspection of preparation for delivery. The preservation, packaging, packing, and marking shall be inspected to determine compliance with the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Preservation, packaging, and packing shall be in accordance with MIL-STD-162, level A, B, or C and type of preparation, as specified (see 6.2).

6. NOTES

6.1 Intended use. The MHU-124/E aerial stores lift truck (electric) is intended for inside use in the integrated maintenance facility, with controlled environment, to load and unload, transport and position aerial stores onto removable pylons, clipons, launchers, and storage stands.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. In the event it becomes necessary to change the drawings (see 3.2) in order to achieve the performance requirements of the specification, the contractor will immediately notify the contracting officer with recommendations for drawing changes to obtain the required performance. To the extent approved, such changes shall be processed in accordance with the provisions of the changes clause of this contract

c. Location and conditions for preproduction testing (see 4.4)

d. Level of preservation, packaging, and packing (see 5.1)

e. Type of preparation (see 5.1)

6.3 Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Air Force - 11

Preparing activity:
Air Force - 11

Reviewer:
Air Force - 79

Project No. 1450-F011

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:50Z
Check the source to verify that this is the current version before use.

FOLD

4950/TZS
Wright-Patterson AFB, O 45433
OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
DEPARTMENT OF THE AIR FORCE
DoD-318

U.S.MAIL

4950/TZS
Wright-Patterson AFB, Ohio 45433

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:50Z
Check the source to verify that this is the current version before use.

12

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

C16061

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:50Z
Check the source to verify that this is the current version before use.